THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 11: **Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. During a lunar eclipse the Moon takes 1 hour to cross the Earth's umbra. The umbra's width relative to the Moon's diameter is about: 1

(a) about 2.5 times **(b)** 10 times **(c)** half **(d)** equal

2. A student thinks 'the dark side of the Moon is always dark'. This is: 1

(a) incorrect **(b)** correct

(c) correct on new moon only **(d)** correct on full moon only

3. If the Moon revolved around the Earth in the same plane as the Earth's orbit, eclipses would occur: 1

(a) randomly **(b)** only once a year **(c)** every month **(d)** never

4. A shadow stick has a shadow of 0 length at noon on the equator on 21 March. This shows: 1

(a) the Sun is overhead **(b)** no light

(c) it is night **(d)** the Sun is at horizon

5. A student says 'we see the same face of the Moon because it does not rotate'. This is: 1

(a) partly correct **(b)** correct

(c) incorrect **(d)** correct on new moon

6. A lunar eclipse is visible at the same time from: 1

(a) only India

(b) all places where the Moon is above the horizon

(c) only the equator

(d) only near the poles

7. The longest shadow of a stick during the day occurs at: 1

(a) noon **(b)** 1 p.m.

(c) sunrise and sunset **(d)** 11 a.m.

8. A student says 'the Earth is closest to the Sun in summer in the northern hemisphere'. This is: 1

(a) correct only in June **(b)** correct

(c) correct only in July **(d)** incorrect

9. If the Moon were closer to the Earth, a total solar eclipse would be: 1

(a) unchanged **(b)** impossible **(c)** rarer **(d)** more common

10. A person in Delhi finds the shadow at noon shorter in June than in December because: 1

(a) day is shorter **(b)** the stick is longer
(c) the Sun is higher in the sky in June **(d)** the Sun is lower

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the shadow of a vertical stick trace a curve through the day? 2

12. Explain the phrase 'It is midnight in India when it is noon in some other places'. 2

13. Explain why the Earth must revolve for seasons but also be tilted. 2

14. Why can a total solar eclipse be seen only from certain places? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A shadow stick 1 m long casts a shadow of 0.5 m at noon in summer and 1.5 m in winter. Compare the Sun's height and explain. 3

16. Describe two methods of finding true north using the Sun or the Pole star. 3

17. Explain with a reason why a place on the equator never experiences polar day. 3

OR

Explain why the number of leap years in 400 years is 97 not 100. 3

18. Explain what causes the sky to turn dark during a total solar eclipse. 3

OR

Why do hill stations have cooler weather than plains even in summer? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss how the combined motions of the Earth and the Moon produce day, month, year, phases and eclipses. 5

OR

Evaluate whether the Moon's phases can be explained by Earth's shadow, with a simple experiment using a ball and lamp. 5

20. A calendar group wants to make a calendar for 8 years including 2 leap years. Explain how to count days and the structure of months. 5

OR

Design a model with a globe, a ball and a lamp to show a solar and a lunar eclipse. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND